PAYSECURE GATEWAY PRIVATE LIMITED

Multi Region Disaster Recovery Architecture

Decision design, operating model, compliance mapping, and recovery runbooks

Prepared for client review | India regions: Mumbai (ap-south-1) and Hyderabad (ap-south-2)

Design baseline: 3.2 million transactions/day | INR 500 crore/day | 45,000 merchants | 1,200 peak TPS

Target: 99.99% service availability, RPO < 60 seconds, RTO < 5 minutes by Q3 2026

CONFIDENTIAL - Fictional PaySecure scenario. This is a target architecture and decision package, not evidence of a deployed or tested production system.

Executive decision

Recommend a warm-standby, active-passive payment write path across Mumbai and Hyderabad. Keep both regions deployed and recovery-ready, but maintain one authoritative writer for payment state. This is the safest fit for financial correctness, the eight-person platform team, and the RPO objective. Active-active ingress/stateless reads may be considered later; dual-region writes to authoritative payment state are not recommended.

Schedule status as of 20 October 2026: the requested Q3 2026 target date has passed. Treat the design as an overdue recovery objective: establish the actual delivery status immediately, agree a revised board-approved date and interim risk controls, then sequence the acceptance gates below.

Decision	Recommendation	Reason
Region pair	Mumbai ap-south-1 primary; Hyderabad ap-south-2 recovery	Both are Indian regions; validate service parity, quotas, connectivity and contracts.
Operating pattern	Warm standby, active-passive for writes	Avoid concurrent payment/settlement authorities; pre-provision recovery capacity.
Database	Aurora PostgreSQL Global Database; one writer	Cross-region replication is asynchronous; data loss depends on lag at failure.
Availability	99.99% monthly SLO	Approx. 52.6 minutes/year allowed downtime; measure end-to-end.
Recovery	RPO <60 sec; RTO <5 min	Acceptance targets, not service guarantees; prove with repeated production-like drills.
Traffic control	Route 53 failover plus ARC routing controls	DNS is one signal; existing connections and resolver caches persist.

Scope

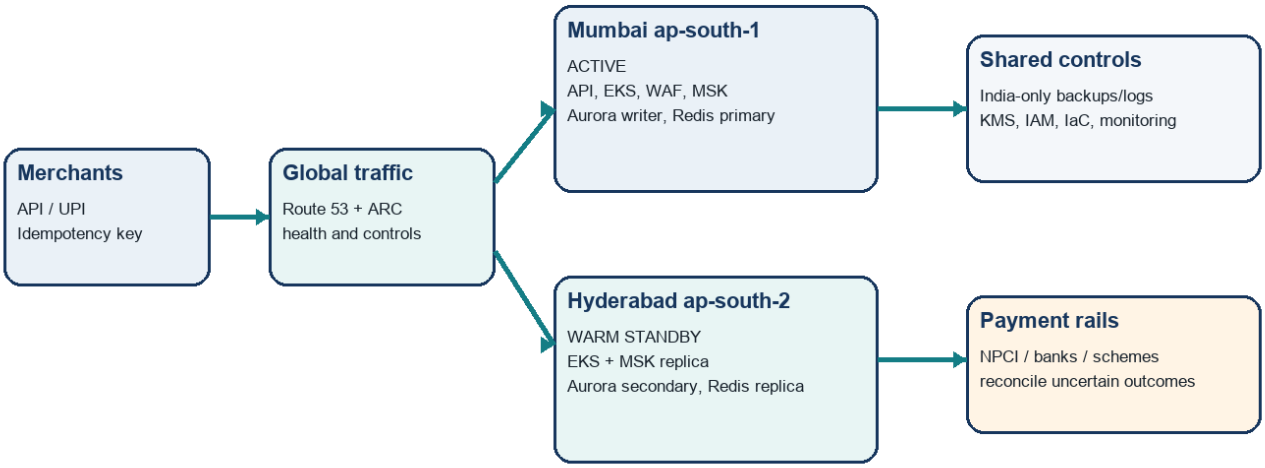
- Architecture and failover sequence
- Replication strategy and limits for Aurora PostgreSQL, DynamoDB, ElastiCache Redis and Kafka/MSK
- Twelve operator runbooks
- Decision mapping to RBI, PCI DSS, NPCI/UPI and Indian payment-data localisation
- Five-stakeholder simulated Business Continuity Review Board

1. Business and resilience basis

Measure	Value / implication
Current availability	99.92% (scenario input), approx. 7h annual downtime.
Target availability	99.99%, approx. 52m 34s/year; about 8x less downtime.
Daily transactions	3.2m/day = approx. 37 average TPS; design to 1,200 stated peak TPS.
Average transaction value	INR 500 crore / 3.2m = approx. INR 15,625 arithmetic mean only.
DR capacity	Benchmark Hyderabad for >=1,560 TPS (1,200 + 30% headroom), including DB/MSK/partner quotas.
RPO	<60 sec for durable payment/ledger records; cache is rebuildable and excluded.
RTO	<5 min to restore critical payment journey, not just DNS or pod readiness.

Critical journey: authenticated merchant request to durable, idempotent accepted/declined outcome or safe retryable response. A timeout after external authorization is an unknown outcome; reconcile by network reference and idempotency key before retry.

2. Architecture overview



Normal operation: Mumbai accepts writes. Hyderabad receives replicated state and runs a warm service tier. During disaster, traffic moves only after primary fencing and recovery-point assessment. DNS TTL does not invalidate cached answers or existing connections.

Layer	Mumbai active	Hyderabad recovery	Control / safety
Edge/API	WAF, ALB, ingress	Equivalent warm stack	Health checks + ARC routing; synthetic durable write tests.
Compute	EKS across >=3 AZs	Deployed; DR baseline nodes warm	Pinned images, same schema, regional secrets and quotas.
Payment state	Aurora writer	Global DB secondary	Single writer, fencing, lag gate, reconciliation.
DynamoDB	Classified regional/global tables	Replicas/endpoints ready	Do not use conflict resolution for balances/settlement.
Cache	Redis primary Multi-AZ	Global Datastore secondary	Never authoritative; protect DB during warm-up.
Events	MSK cluster	MSK cluster + Replicator	At-least-once, idempotent consumers and offset plan.
Security	Regional KMS/logging/IAM	Independent keys and logs	MFA, CloudTrail, immutable India backups.

3. Active-active versus active-passive

Criterion	Active-active	Warm active-passive (selected)
Correctness	Requires conflict control and global uniqueness; partition can create two writers.	One writer; clear ledger ownership and failover.
Latency	Good for local stateless/read traffic; cross-region writes add latency.	Mumbai normal latency unchanged; Hyderabad ready, not writing.
Operations	Two production footprints, more complex partial failures.	Explicit promotion and simpler operator decision.
Cost	Full duplicated active capacity.	Warm capacity costs less than full active duplication.
Decision	Possible later for stateless APIs after proof.	Chosen for financial write path and settlement.

4. Component replication and recovery

Aurora PostgreSQL

- Aurora Global Database, primary in ap-south-1 and secondary in ap-south-2; asynchronous storage replication, not synchronous cross-region commits. Monitor AuroraGlobalDBReplicationLag/RPO lag.
- Keep a secondary DB instance online; pin engine, extensions, parameter/schema versions, TLS, IAM auth, secrets and network policy.
- Failover: stop/fence Mumbai writes; promote Hyderabad only on IC declaration; confirm one writable primary; refresh application endpoints; run payment/reversal/idempotent-retry probes.
- RPO <60s is monitored, not guaranteed. Failover can lose data since the secondary applied point. Use idempotency and external rail reconciliation. Never auto-failback.

DynamoDB

- Classify tables before Global Tables. MREC can serve suitable non-ledger metadata only with asynchronous conflict handling. Financial truth stays in Aurora.
- Use conditional writes, immutable IDs, PITR, deletion protection, encryption and ReplicationLatency alarms. Do not use last-writer-wins for balances, settlement, refunds or authorization.
- MRSC is not the default: it needs three regions/replicas or a witness, with topology and latency constraints. Confirm current supported geography before design use.

ElastiCache Redis

- Multi-AZ within each region and Global Datastore Mumbai primary to Hyderabad secondary; cross-region replication asynchronous and regional promotion manual.
- Cache only rebuildable values. Never store the only copy of balance, idempotency decision or settlement outcome.
- On promotion, expect cold keys; use request coalescing, bounded fills, DB protection and gradual traffic ramp.

Kafka / MSK

- Independent MSK clusters with MSK Replicator, required topics, ACLs, schema compatibility, retention, quotas and offset sync where supported. Replication is at-least-once and asynchronous.
- Every financial event has immutable event_id, payment_id, source_region, schema_version and idempotency key. Consumers deduplicate; use transactional outbox/inbox.
- At failover, stop/fence producers where possible, record lag/high-water marks, choose target offsets, enable consumers and reconcile from Aurora outbox/partner acknowledgements. Kafka is not the ledger.

5. Five-minute failover objective

Elapsed target	Action / owner	Gate
0:00-0:30	Detect via independent synthetic, API error and dependency alarms; on-call bridges.	Correlated probes from >=3 networks.
0:30-1:00	IC declares; freeze deploys and fence writes.	Two-person approval except documented emergency override.
1:00-2:45	Check lag; promote Aurora; activate DR event/cache endpoints as required.	Single writer; record last applied point.
2:45-3:45	Canary at 5-10%; routing via ARC/Route 53; refresh pools.	Payment, reversal, retry, reconciliation pass.
3:45-5:00	Ramp traffic; Hyderabad capacity >=1,560 tested TPS.	No duplicates; queues, p99 and issuer error rates within limits.

These elapsed times are engineering goals. Aurora promotion can take minutes; DNS caches, existing sessions, provider APIs, scheme behavior and dependencies can extend recovery. Accept the RTO only after repeated measured end-to-end drills.

6. Security, locality and recoverability

Control	Implementation
India locality	Payment messages, records, sensitive logs, backups, replicas and recovery artifacts stay in India. Inventory telemetry and vendor-support paths; redact PAN/auth data.
PCI scope	Replicate required CDE only; preserve segmentation, WAF, access logs, scanning and evidence in DR; have QSA validate scope.
Encryption	Regional KMS keys and tested decrypt path; controlled grants; do not assume key replica alone makes secrets usable.
Identity	Separate regional roles, least privilege, MFA, short-lived credentials and two-person promotion/routing approval.
Cyber recovery	PITR, immutable in-India backup, account separation, deletion protection, clean-room restore; replication can copy corruption.
Observability	Independent regional logs/metrics/alerts; monitor DB/Kafka/cache lag, queue age, business success, and redact payment data.
Change discipline	Expand-contract schema; deploy compatible versions to both regions; freeze risky changes during incidents.

7. Regulatory and control traceability

This is a control mapping, not a legal opinion or compliance attestation. Applicability depends on PaySecure's exact authorisation and role. NPCI participant standards can be member-portal or contract controlled. The source brief's universal RBI four-hour RTO and NPCI 99.95% uptime/300 ms claims are not treated as verified binding requirements.

Decision	RBI / governance	PCI DSS v4.x	NPCI / UPI	Locality
Regional DR site	BCP, risk governance, service-provider oversight; confirm applicable PA/PSO clauses.	12.10.1 response/recovery; 12.10.2 testing.	DR endpoint/certification terms from current participant agreement.	Both regions Indian.
Single writer + fencing	Operational risk, reconciliation and access controls.	12.10 containment; Req. 7/8 access.	Prevent duplicate participant messages; follow transaction/reversal rules.	Avoid divergent payment data.
Aurora lag gate	Document objectives, test recovery, retain audit evidence.	12.10 recovery/test; Req. 10 logs.	Preserve transaction refs and reconciliation.	Replicas/backups in India.
DynamoDB conflict policy	Data integrity, vendor governance, audit.	Req. 3 stored data; Req. 10 logging.	Avoid changing transaction semantics.	Only India replicas.
Redis non-authoritative	Resilience and access controls.	Req. 3 if account data; Req. 7/8.	Rebuild cache without altering transaction state.	India storage; minimize sensitive data.
Kafka replication	BCP, incident response, reconciliation.	12.10; Req. 10 logging.	Idempotent reprocessing and business-required order.	Topics/offsets/backups in India.
Traffic cutover	BCP, change controls, tested incident roles.	12.10 roles/comms/recovery; Req. 1 network.	Approved endpoints and partner connectivity.	No payment content in DNS/telemetry.
CDE segmentation/ encryption	Applicable RBI security controls and board policy.	Req. 1,3,4,7,8,10.	Protect UPI credentials/messages and keys.	Key/log/storage paths remain in India.
Backup/restore	Continuity, records, audit and provider access.	Req. 3 backup; Req. 12 response.	Retain dispute/reconciliation records per applicable terms.	Immutable copy in India.

Decision	RBI / governance	PCI DSS v4.x	NPCI / UPI	Locality
Exercise cadence	Board-approved BCP and applicable regulator evidence.	12.10.2 at least annual test; 12.10.3 24/7 response.	Controlled partner certification tests.	Synthetic/masked drill data.
SLOs/communications	Board objectives; verify exact reporting obligations.	12.10 incident response.	Use current uptime/latency/notice terms, not inferred universal thresholds.	Monitor residency and vendor access.

8. Twelve disaster recovery runbooks

Common prerequisites: on-call has approved regional access with MFA; incident bridge works; AWS IDs/endpoints are in a controlled inventory; probes are synthetic; UTC is synchronized. Record operator, approver, timestamp, before/after state. Replace variables from inventory; never paste secrets into shell history.

RB01 Complete Mumbai regional failure

Trigger: independent synthetic fails >=60 sec plus regional impairment, or IC declares.

Execution: Open P1 bridge/freeze deploys; confirm Hyderabad health/replication lag; fence Mumbai writes; promote Aurora secondary; verify one writer; promote Redis only if needed; inspect Kafka lag/high-water marks and enable DR consumers; set ARC/Route 53 routing; refresh pools; test payment, reversal and idempotent retry; ramp 10/50/100%.

Owners and exit: IC, DB/app/payment leads. Capture declaration, lag, writer ARN, Kafka offsets and references. Exit: critical journey healthy, gap reconciled. No auto-failback.

RB02 Partial regional or AZ degradation

Trigger: AZ/dependency alarm with Mumbai partially healthy.

Execution: Map blast radius; check multi-AZ health; keep regional promotion disabled unless journey fails or IC declares; drain unhealthy targets and scale healthy AZs; check DB writer/Kafka ISR; invoke RB01 if objective breach is likely.

Owners and exit: IC/service owner. Exit: error budget stabilizes and AZ redundancy restored. Avoid promotion for control-plane-only alarm.

RB03 Aurora writer or regional DB failure

Trigger: writer unavailable >30 sec, severe DB errors or region loss.

Execution: Check cluster/replica lag; suppress unsafe retries; allow same-region Multi-AZ recovery first; for regional failure fence old writer; run `aws rds failover-global-cluster --global-cluster-identifier "$GLOBAL_CLUSTER" --target-db-cluster-identifier "$DR_CLUSTER_ARN"`; poll `aws rds describe-global-clusters`; confirm sole writer; refresh app endpoint and smoke test.

Owners and exit: DB lead. Record lag, applied point, output and transaction IDs. Exit: one writer and ledger checks pass; if old writer ambiguous, hold writes and escalate.

RB04 Database split-brain / dual-writer

Trigger: two endpoints accept writes or role/ledger ambiguity.

Execution: Close payment admission; stop settlement consumers; fence both sides at app/IAM/network; preserve logs and snapshots; inspect global membership, DB read-only probes, ledger/outbox and external refs; IC+DB lead appoint one authority; isolate other; reconcile duplicates/missing events before reopening.

Owners and exit: IC, DB, fraud/risk, finance. Exit: one writer and signed reconciliation; preserve forensic evidence.

RB05 Kafka partition / replication lag

Trigger: producer errors, under-replicated partitions, consumer lag or replicator lag breach.

Execution: Identify topics/partitions; pause noncritical consumers; check broker/ISR/storage/network/Replicator; repair in place if region healthy; on regional loss use RB01, inspect high-water marks and translated offsets, restart without skipping and allow duplicates; replay with dedup; compare outbox and partner acknowledgements.

Owners and exit: Streaming/app lead. Exit: lag inside threshold and no unexplained event IDs. Offset sync is not exactly-once.

RB06 DynamoDB delay, throttling or conflict

Trigger: ReplicationLatency >30 sec for two periods, throttling or conflict spike.

Execution: Identify table/mode/write region; inspect latency, throttles, capacity and status; for MREC halt conflicting key writes and route mutations to home region; never resolve financial state via last-writer-wins; check quotas before scaling; compare item versions and rebuild derived data from authority.

Owners and exit: DynamoDB owner. Exit: replication within budget and deterministic conflicts resolved.

RB07 Redis outage / regional promotion / cold cache

Trigger: primary unavailable, region loss or cache lag risk.

Execution: Confirm cache-only key families; allow same-region Multi-AZ recovery; for region loss obtain IC approval and promote Global Datastore secondary using approved AWS operation; verify role/endpoint; update clients and connection pools; enable request coalescing, fill limits and DB protection; ramp warm-up.

Owners and exit: Cache/app lead. Exit: errors stable and DB safe. Gate payment if non-reconstructable state found.

RB08 DNS / ARC routing-control problem

Trigger: route changes fail, stale endpoint persists or health checks disagree.

Execution: Check endpoint from multiple networks, Route 53 health and ARC safety state; avoid repeated toggles; if control-plane blocked, use approved alternate endpoint and merchant instructions; account for resolver caches/long sessions; verify with merchant-like resolvers.

Owners and exit: Network lead + IC. Exit: traffic reaches chosen region and no split-brain; record DNS tail separately.

RB09 Peak-load failover / festival surge

Trigger: >=80% capacity, queue age rising, or surge coincident with failover.

Execution: Apply priority/admission policy and pause noncritical batches; confirm Hyderabad benchmark >=1,560 TPS plus dependency quotas; enforce per-merchant limits; ramp 5/10/25/50/100% with 60-sec holds; scale within tested bounds; shed noncritical load if needed.

Owners and exit: IC, capacity and merchant ops. Exit: 15 min at target TPS, safe queues, no duplicates and headroom.

RB10 Corruption / ransomware / malicious deletion

Trigger: integrity alarm, destructive API event or malware indicator.

Execution: Declare security P1; isolate identities/write paths; preserve CloudTrail/DB/Kafka/network evidence; do not replicate suspect state; restore clean point from immutable India backup into isolated account/VPC; validate schema/balances/malware/refs; cut over only after security and business approval.

Owners and exit: Security IC, DB, finance. Exit: clean point attested, reporting assessed, evidence preserved; RTO may be exceeded for integrity.

RB11 NPCI/UPI, bank or scheme connectivity incident

Trigger: PaySecure region healthy but rail/participant routes degrade.

Execution: Compare independent non-UPI probes; check partner circuit, mTLS, allowlist, DNS and status; do not promote region unless PaySecure path caused; use channel circuit breaker and safe retryable unknown outcome; query by network reference and reconcile; notify per current contract.

Owners and exit: Payment operations/partner lead. Exit: rail recovers or approved route active; unknowns reconciled.

RB12 Planned failback to Mumbai

Trigger: Mumbai rebuilt, clean, caught up; window and settlement owners approve.

Execution: Verify software/schema/security parity; rejoin as Aurora secondary; verify lag stable; rebuild reverse Kafka replication and validate offsets/dedup; warm Redis; reconcile; run synthetic/refund/partner/load tests; freeze writes if tested process requires; perform managed Aurora switchover; ramp 5/25/50/100%, retain rollback to Hyderabad.

Owners and exit: IC, change, DB/stream/app/finance leads. Exit: Mumbai sole writer, Hyderabad standby, SLO and evidence complete. Never fail back to meet a calendar date.

9. Business Continuity Review Board simulation

Prepared challenge prompts; these are not actual stakeholder statements or approvals.

Stakeholder	Challenge	Response / evidence	Disposition
CTO	Can a five-minute RTO hold if Aurora promotion takes minutes and DNS caches?	Warm instances, ARC plus DNS, measured end-to-end drills; no external commitment until repeated proof.	Conditional; RTO drill is release gate.
CRO	What is worst-case exposure at 45-sec lag with Kafka replay?	At 1,200 TPS, 45 sec is 54,000 attempts theoretical, not a loss estimate. Use one-writer fence, idempotency, outbox and rail reconciliation.	Require risk appetite and finance sign-off.
Compliance Head	How prove locality and obligations?	Region allowlist, inventory of flows/support, India backups, residency evidence and current role-specific legal/NPCI review; QSA validates CDE.	Conditional compliance evidence.
VP Engineering	Can 8 platform engineers run active-active?	Selected warm standby simplifies authority; automate evidence, retain human promotion gate, exercise on-call.	Approve warm active-passive; reject unproven dual write.
External auditor	Incremental cost and value?	SKU model includes standby, replication, transfer, monitoring, retention; quote after sizing and set cost ceiling.	Require FinOps baseline and post-audit.

Simulated board outcome: conditional endorsement for implementation planning pending service/quota validation, role-specific legal/NPCI confirmation, measured <60-sec RPO/<5-min RTO and 1,560 TPS drill, finance reconciliation sign-off, QSA review, and FinOps estimate. No simulated approval is a real customer or regulatory decision.

10. Delivery roadmap and acceptance gates

Phase	Work	Exit evidence
Discovery (2-3 wk)	Data-flow, service/partner, legal-role and critical journey inventory.	Approved workload map, SLO, RACI, partner validation.
Foundations (3-5 wk)	ap-south-2 landing zone, network, KMS, IAM, EKS, logs, quotas, artifacts.	Policy/security review; India-only payment data; DR deploy from IaC.
Data plane (4-6 wk)	Aurora, DynamoDB classification, Redis, MSK; idempotency/outbox/reconciliation.	Measured lags, integrity, PITR and failover rehearsals.
Application (3-5 wk)	Region-aware endpoints, safe retries, load shedding, replay, notifications.	Compatibility and duplicate/unknown outcome tests; >=1,560 TPS DR test.
DR drills (2-3 wk)	Regional, split brain, Kafka, peak load, failback.	Three successful end-to-end drills meeting defined objectives.
Production by Q3 2026	Staged cutover, change approval, watch and evidence retention.	Signed go/no-go, rollback, compliance and board acceptance.

11. Risks, assumptions and open decisions

- The three objectives require measurement. Aurora and streaming replication are async; recovery timing varies.
- Confirm exact Aurora engine/API promotion semantics, secondary readiness, quota and connection refresh in nonproduction.
- Validate ap-south-2 service parity and quotas at build time using a region feature matrix.
- Confirm NPCI/sponsor-bank endpoint registration, allowlists, mTLS, callbacks, certification, notifications and status inquiry.
- Confirm current RBI authorisation class, binding Master Direction/circulars and reporting duties.
- Define which durable states fall inside the 60-second RPO: payment decision, ledger, outbox and settlement instruction.
- Define SLI window, maintenance exclusions and upstream attribution; publish end-to-end and platform success rates.
- Incremental cost cannot be asserted without right-sizing, throughput, retention, transfer and negotiated discounts.

Appendix A User requirement coverage

Requirement	Where addressed
Compare/select active-active or passive	Section 3 compares options and selects warm active-passive for payment writes.
AWS Indian regions	Mumbai and Hyderabad; locality and validation gates.
Four named data technologies	Section 4 gives specific replication, lag, promotion, integrity and limits.
12 production runbooks	RB01-RB12 include triggers, actions, owners and exit evidence; populate controlled inventory and rehearse commands.
Required failure scenarios	Region RB01, split brain RB04, Kafka RB05, peak load RB09; remaining incidents covered.
Regulatory map	Section 7 maps major decisions to four named frameworks and flags applicability.
Five-stakeholder board	Section 9 includes CTO, CRO, Compliance, VP Engineering and auditor.

Appendix B Source references

Primary sources consulted. Confirm operative versions and entity-specific application with counsel, compliance, QSA, AWS, and PaySecure counterparts before adoption.

RBI - Storage of Payment System Data, 6 April 2018

RBI - Storage of Payment System Data FAQ

RBI - Payment Aggregators and Payment Gateways directions

RBI - Digital Payment Security Controls

PCI SSC - PCI DSS v4.0.1 update

PCI SSC - Requirement 12.10 sample official text

NPCI - UPI circulars

AWS - Aurora Global Database DR

AWS - DynamoDB Global Tables

AWS - DynamoDB consistency / region support

AWS - ElastiCache Global Datastore

AWS - ElastiCache regional promotion

AWS - MSK cross-region replication

AWS - MSK Replicator delivery considerations

AWS - Route 53 health check/failover

AWS - Route 53 active/passive concepts